Theory Assignment-1

Q1. Explain the HTTP Request–Response Cycle with a suitable diagram. Describe the major steps involved when a client (browser) requests a resource from a web server and receives a response.

Ans:

The **HTTP Request–Response Cycle** is the process by which a browser (client) communicates with a web server to request and receive a web resource.

Major Steps

1. **Request:** The user enters a URL, and the browser creates an HTTP request.
2. **Send:** The request is sent to the web server.
3. **Processing:** The server processes the request and finds or generates the required resource.
4. **Response:** The server sends an HTTP response containing a status code and requested data.
5. **Rendering:** The browser receives the response and displays the webpage/resource.

Example:
GET /index.html → Server processes it → 200 OK + HTML page → Browser displays the page.

Q2. Briefly explain the purpose and key features of the following web technologies:

1. HTML
2. CSS
3. JavaScript
4. Bootstrap
5. jQuery

6. React.js

Ans:-

1. HTML

HTML (HyperText Markup Language) is used to create the structure of web pages.
Key features: headings, paragraphs, links, images, forms, tables, and semantic elements.

2. CSS

CSS (Cascading Style Sheets) is used to design and style web pages.
Key features: colors, fonts, layouts, spacing, animations, and responsive design.

3. JavaScript

JavaScript is a scripting language used to make web pages interactive and dynamic.
Key features: event handling, form validation, DOM manipulation, animations, and API calls.

4. Bootstrap

Bootstrap is a front-end CSS framework used to create responsive and attractive websites quickly.
Key features: grids, buttons, forms, navigation bars, cards, and responsive components.

5. jQuery

jQuery is a JavaScript library that simplifies DOM manipulation and event handling.
Key features: easy selectors, animations, event handling, and AJAX requests.

6. React.js

React.js is a JavaScript library used to build interactive user interfaces, especially single-page applications.
Key features: reusable components, Virtual DOM, JSX, state management, and efficient UI updates.

Q3. What are client-side technologies and server-side technologies? Explain the difference between them with suitable examples.

Ans:- Client-side technologies are technologies that run on the user's web browser. They are mainly used to create the appearance and interactivity of a website.
Examples: HTML, CSS, JavaScript, React.js.

Server-side technologies run on the web server. They handle business logic, database operations, authentication, and processing of requests.
Examples: Python, PHP, Java, Node.js, ASP.NET.

Difference

1. Client-side code runs in the browser, while server-side code runs on the server.
2. Client-side is mainly used for UI and user interaction, while server-side handles data processing and business logic.
3. Client-side technologies include HTML, CSS, and JavaScript, while server-side technologies include Python, PHP, Java, and Node.js.

Example: In a login system, JavaScript can validate the form on the client side, while the server checks the username and password against a database.

Q4. What are block-level elements in HTML? Explain their characteristics and provide at least three examples.

Ans:-

Block-Level Elements in HTML

Block-level elements are HTML elements that start on a new line and usually occupy the full available width of their parent element.

Characteristics:

- Start on a new line.
- Usually take up the full width available.
- Can contain other block-level and inline elements.
- Can have width, height, margin, and padding applied.

Examples:

1. – Defines a general block/container.
2. <p> – Defines a paragraph.
3. <h1> to <h6> – Defines headings.
4. <section> – Defines a section of a webpage.

Q5. What are semantic elements in HTML? Explain their importance in web development and provide suitable examples.

Ans:-

Semantic Elements in HTML

Semantic elements are HTML elements that clearly describe the meaning and purpose of their content.

Importance:

- Improve code readability and structure.
- Help search engines understand webpage content.
- Improve accessibility for screen readers.
- Make websites easier to maintain and develop.

Examples:

- <header> – Defines the header of a page or section.
- <nav> – Defines navigation links.
- <main> – Defines the main content.
- <section> – Defines a section of content.
- <article> – Defines independent content.
- <footer> – Defines the footer.

Example:

```
<header>My Website</header>
<nav>Home | About | Contact</nav>
<main>
  <article>
    <h1>Welcome</h1>
    <p>This is my webpage.</p>
  </article>
</main>
<footer>Copyright 2026</footer>
```

In short: Semantic elements make HTML meaningful, structured, accessible, and SEO-friendly.

Q6. Write HTML code to create the following table: Sr. No. Name Designation Department 1 John Manager Sales 2 Smith Sr. Manager Marketing 3 Jane Manager HR The table should include appropriate table headings, rows, and columns.

Ans:-

```
<!DOCTYPE html>
<html>
<head>
   <title>Employee Table</title>
</head>
<body>
<table border="1">
   <tr>
      <th>Sr. No.</th>
      <th>Name</th>
      <th>Designation</th>
      <th>Department</th>
   </tr>
   <tr>
      <td>1</td>
      <td>John</td>
      <td>Manager</td>
      <td>Sales</td>
   </tr>
   <tr>
      <td>2</td>
      <td>Smith</td>
      <td>Sr. Manager</td>
      <td>Marketing</td>
   </tr>
```

```
    <tr>
        <td>3</td>
        <td>Jane</td>
        <td>Manager</td>
        <td>HR</td>
    </tr>
</table>
</body>
</html>
```

Q7. Write HTML code to create the following list using an appropriate combination of ordered/unordered lists:

- River
    - Ganga
    - Yamuna
    - Brahmaputra
    - Narmada

Ans:-

```
<!DOCTYPE html>
<html>
<body>
<ol>
    <li>River
        <ul>
            <li>Ganga</li>
            <li>Yamuna</li>
            <li>Brahmaputra</li>
            <li>Narmada</li>
        </ul>
    </li>
</ol>
```

```
</body>

</html>
```

Q8. Explain the different types of CSS used for applying styles to an HTML document. Describe each type with a suitable example.

Ans:-

Types of CSS

There are three types of CSS used to apply styles to an HTML document:

1. Inline CSS

CSS is written directly inside the HTML element using the style attribute.

Example:

```
<p style="color: blue;">Hello World</p>
```

2. Internal CSS

CSS is written inside the <style> tag in the <head> section of the HTML document.

Example:

```
<head>
    <style>
        p {
            color: blue;}
    </style>
</head>
```

3. External CSS

CSS is written in a separate .css file and linked to the HTML document using the <link> tag.

HTML:

```
<link rel="stylesheet" href="style.css">
```

style.css:

```
p {
    color: blue;}
```

In short: Inline CSS is used for a single element, Internal CSS for one webpage, and External CSS for multiple webpages.

Q9. Differentiate between HTML, CSS, and JavaScript based on their role in developing a web page. Explain what each technology contributes to a website.

Ans:-

HTML, CSS, and JavaScript are the three main technologies used to develop web pages.

1. HTML (Structure)
HTML defines the structure and content of a webpage. It adds headings, paragraphs, images, links, forms, tables, etc.
Example: <h1>Welcome</h1>

2. CSS (Presentation)
CSS is used to style and design the webpage. It controls colors, fonts, spacing, layout, and responsiveness.
Example: h1 { color: blue; }

3. JavaScript (Behavior)
JavaScript adds interactivity and dynamic behavior to a webpage. It handles events, form validation, animations, and dynamic content.
Example: alert("Welcome!");

In short:

HTML → Structure
CSS → Style
JavaScript → Behavior/Interaction

Q10. Answer the following:

a. What is the role of Bootstrap in web development?

b. What is jQuery and why was it widely used in web development?

c. What is React.js and what problem does it solve in frontend development?

Ans:-

a. Role of Bootstrap in Web Development

Bootstrap is a front-end framework used to create responsive and attractive websites quickly. It provides ready-made CSS classes and components such as buttons, forms, navigation bars, cards, and grids.

b. jQuery

jQuery is a JavaScript library that simplifies tasks such as DOM manipulation, event handling, animations, and AJAX requests. It was widely used because it made JavaScript easier, shorter, and more compatible across different browsers.

c. React.js

React.js is a JavaScript library for building user interfaces. It uses reusable components and a Virtual DOM to efficiently update the webpage.

It solves the problem of managing complex and dynamic user interfaces by allowing developers to divide the UI into small, reusable components.